

Data Governance Onboarding Program – INVX

Operationalization – Data Quality Management (continue)

DataX Data Governance Office (Group COE)

13 December 2023



Activities and Plan



✓ DQ Operating shape up

Expectation and what we will achieve



✓ Define and Discover Phase

Introduce Business rule with generic rule automation for CDEs



Diagnose Phase

Conduct Business rules refinement
Illustrate DQ results design and DQ scorecard



Remediate and monitor Phase

Recap DQ Remediation Process



Q&A

If any

Agenda

1

Discover Phase (*continue*)

2

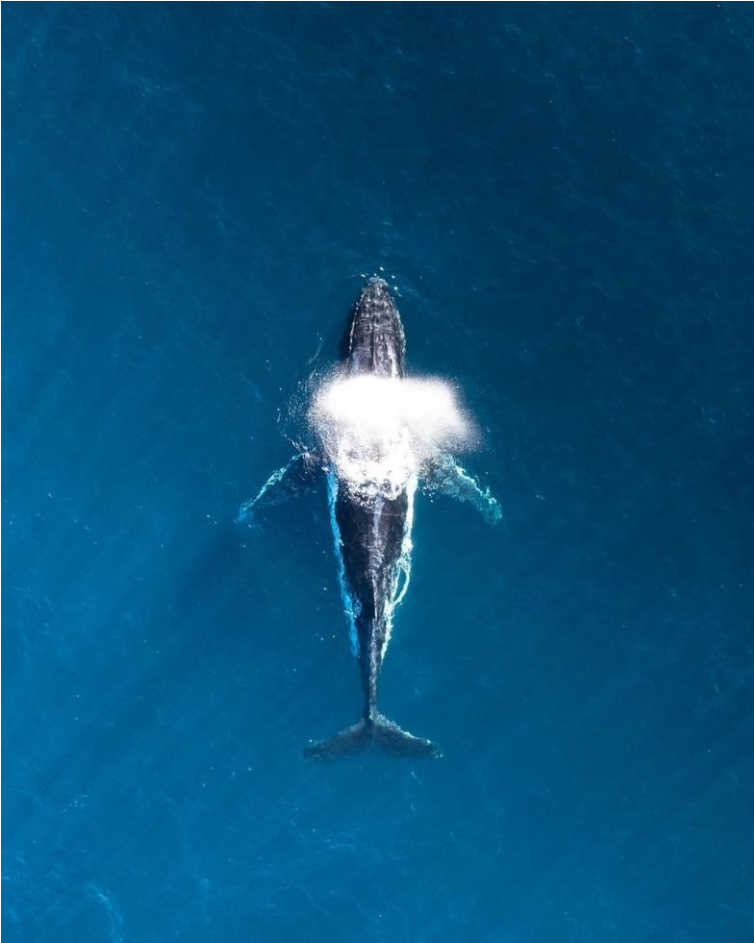
Diagnose Phase

3

Remediate Phase

4

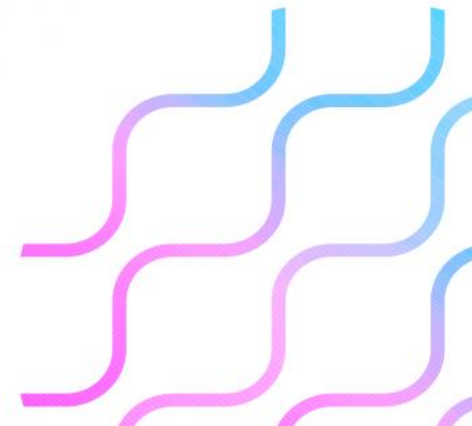
Monitor Phase

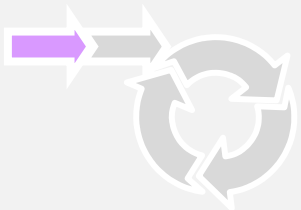


01

Discover Phase (continue)

DQ Business Rules Identification and Refinement





Identify / Maintain Critical Data Elements (CDEs)



Map DQ Dimension

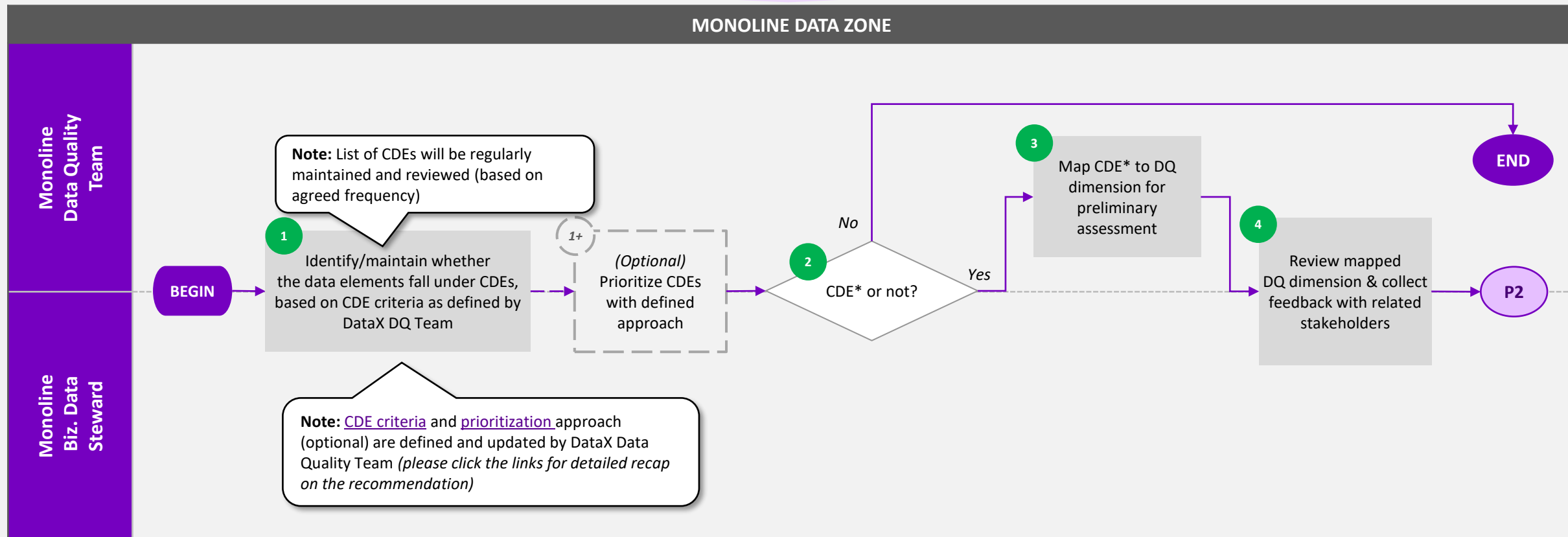


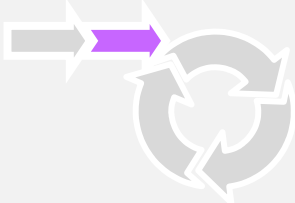
Review with Business Data Steward

Prerequisite(s)

- Data are already provisioned into Monoline Data Zone

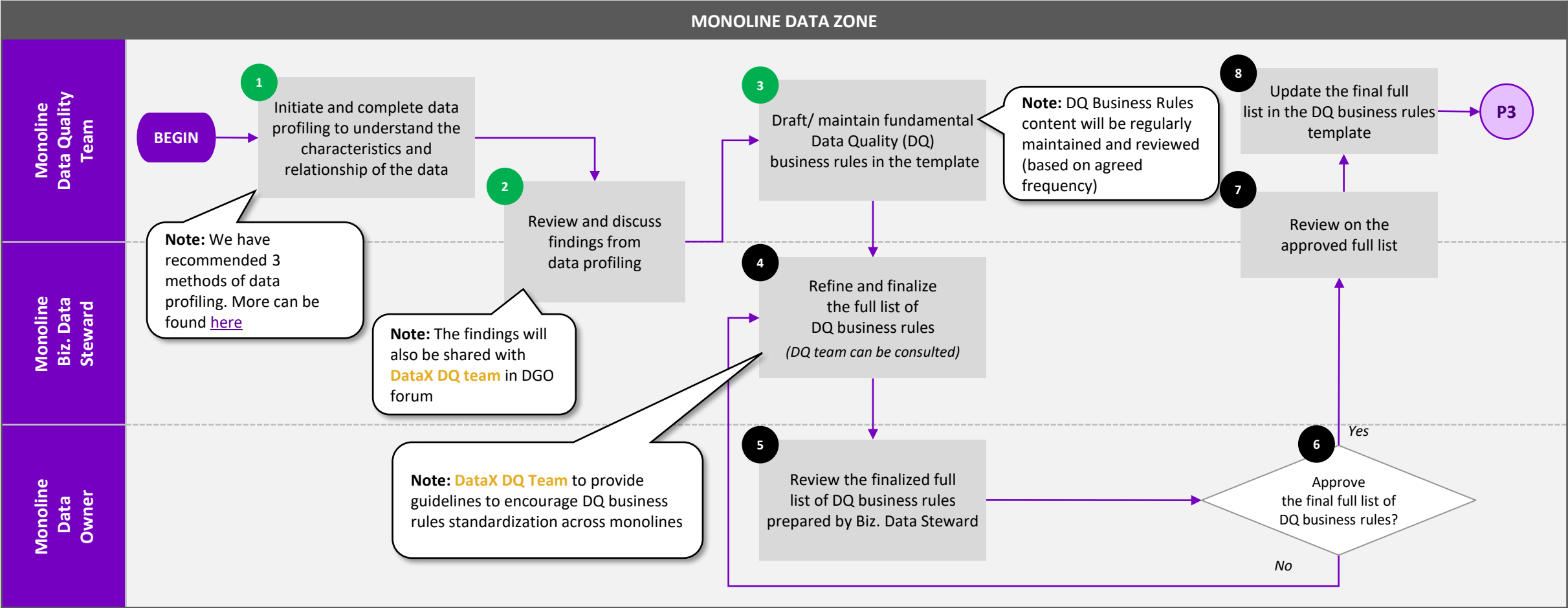
MONOLINE DATA ZONE



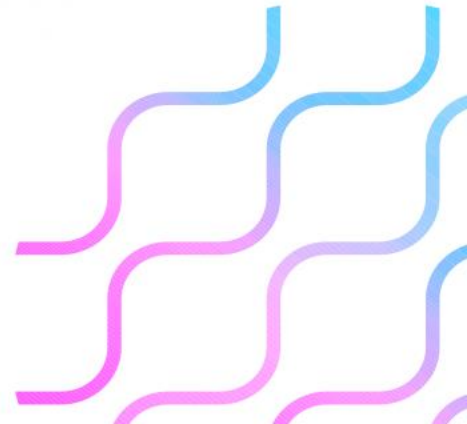


Prerequisite(s)

- CDEs are already defined

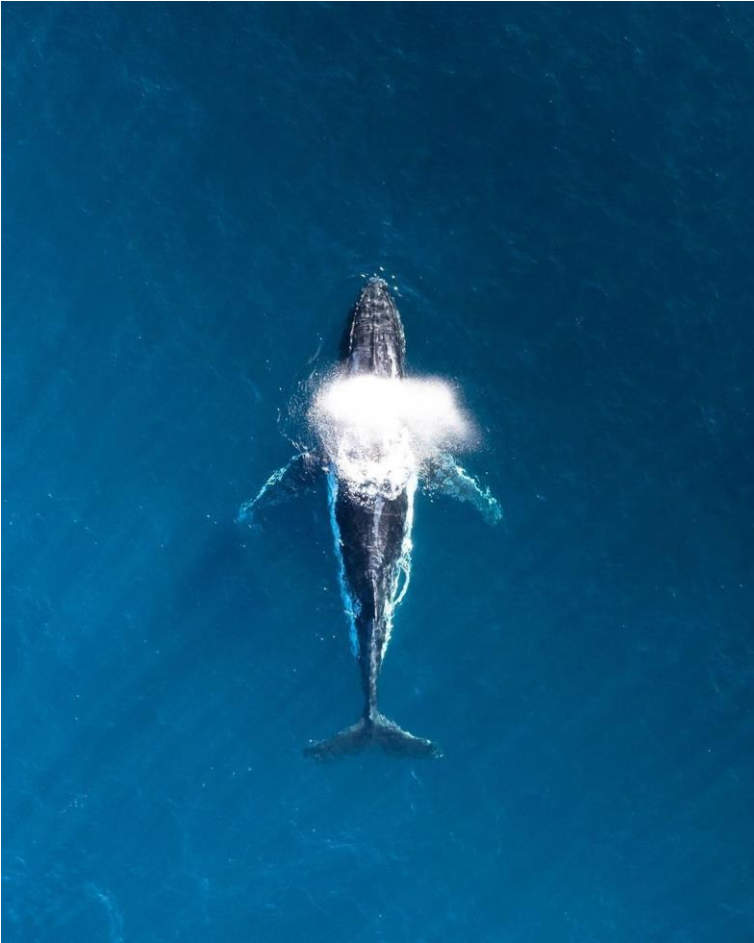


- Walk thru draft version
- Guide to review for finalized version



PHASE	DESCRIPTION
DEFINE / REFINE Process 1: CDE Identification/Maintenance and DQ Dimension Mapping Process	Define / refine foundational DQ components, incl. identification / maintenance of Critical Data Elements (CDEs) and mapping of DQ dimensions
DISCOVER Process 2: DQ Business Rules Identification and Refinement	Profile data, refine and confirm DQ business rules for implementation, and apply global DQ Index Threshold
DIAGNOSE Process 3: DQ Scorecard Generation and Analysis Process	Generate DQ scorecard and conduct high-level analysis to identify DQ issues
REMEDiate Process 4: DQ Issue Analysis Process	Prioritize issues and conduct root-cause and impact analysis for DQ issue
MONITOR Process 5: DQ Issue Remediation Process 6: DQ Scorecard Monitoring	Monitor progress on remediation, as well as continuously monitor DQ scorecard, to ensure improvement of data quality

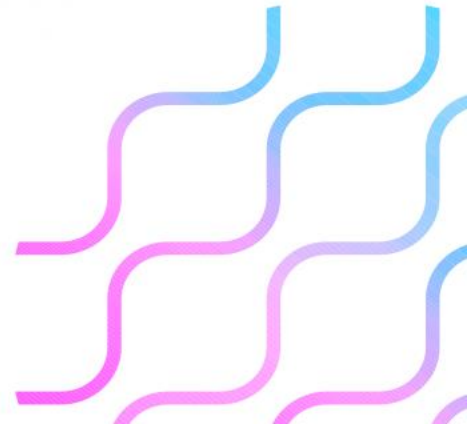
FOCUS FOR TODAY



02

Diagnose Phase

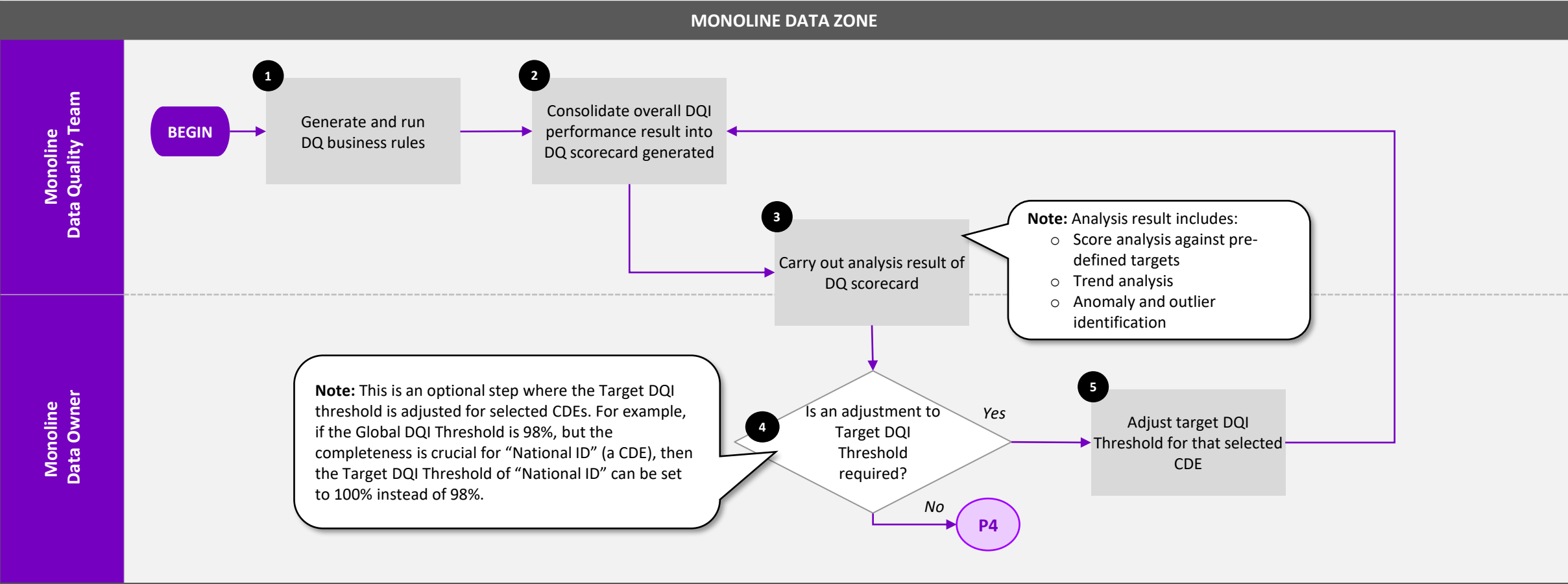
DQ Scorecard Generation and Analysis Process



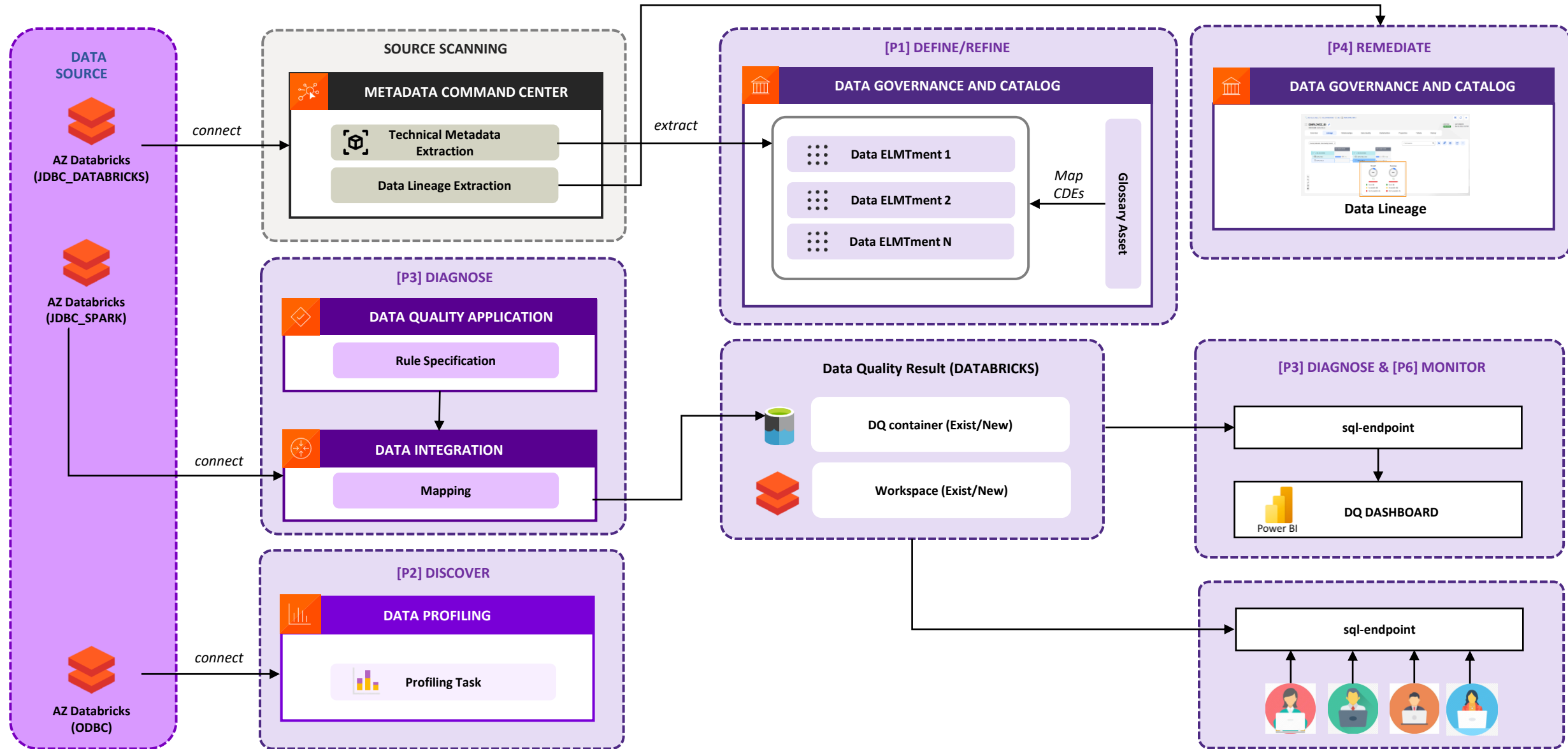
PROCESS 3 (MDZ) TO GENERATE AND ANALYZE DQ SCORECARD

Prerequisite(s)

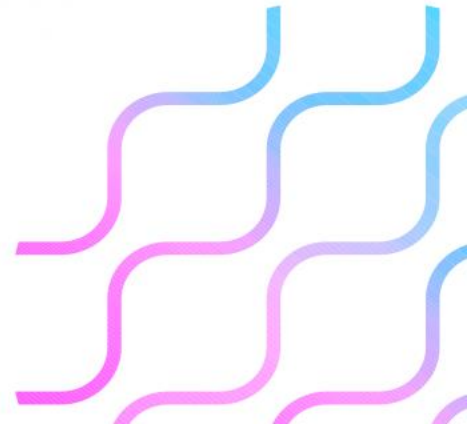
- Business rules for data in MDZ have been defined/ refined and implemented in the tool
- DQ Scorecard in MDZ has been built and updated according to the latest business rules



HIGH LEVEL DATA QUALITY IMPLEMENTATION MODULE



- Live demo using DG Tool





Data Quality Availability & Timeliness Dimension

Data Refresh: 12/12/2023 10:14:03

Start date

End date

28/9/2023

26/11/2023

Data Subject Area

Search

☐ CUSTOMER PROFILE

☐ DEPOSIT

☐ SECURED LOAN

Dataset

Search

☐ core_datax_cust_prfl_cust_basic_aggrgt...

☐ core_scb_dpst_txn_acct_basic_aggrgte...

☐ core_scb_dpst_txn_acct_basic_aggrgte...

☐ core_scb_dpst_txn_acct_basic_aggrgte...

DQ Availability & Timeliness

Reset

Frequency

Daily

Monthly

Status

Anomaly Records

Success

Zero Record

Anomaly Data Table

Dataset name	Fail Zero	Fail Anomaly	Not Started
core_scb_dpst_txn_acct_basic_aggrgte_mnth	2	1	43
core_datax_cust_prfl_cust_basic_aggrgte_day	1		41

Over all DQ Score: **Daily**

88.6228%



Yellow

Fail by Anomaly Records (IQR Formulas)

Green

Success

Red

Fail by zero record

★ Latest Data Date

Not started

1. The job failed 2. It's not the scheduled time yet

Detail Matrix

Dataset name \ Data date	Frequency	Day delay	2023-11-03	2023-10-31	2023-10-28	2023-10-15	2023-10-14	2023-10-13	2023-10-12	2023-10-11	2023-10-10
core_datax_cust_prfl_cust_basic_aggrgte_day	Daily	D-2	0	12727918		12671031	12672290	12672290	12672290	12672290	12672290
core_scb_dpst_txn_acct_basic_aggrgte_day	Daily	D-2	0	10562016		7785093	7429001	7591785	8060591	7989062	8630877
core_scb_dpst_txn_acct_basic_aggrgte_hrs	Daily	D-2	0	24035772		17404947	16615014	17174423	18156173	17957846	19347419
core_scb_dpst_txn_acct_basic_aggrgte_mnth	Daily	D-2		22679060	22020675				22020675	22020675	22020675
solid_datax_cust_prfl_cust_basic_aggrgte_day	Daily	D-2	0	12727918		12671031	12672290	12672290	12672290	12672290	12672290
solid_scb_auto_acct_acct_basic_aggrgte_day	Daily	D-2	0	652718		645633	645546	645472	645393	644933	644570
solid_scb_auto_acct_cust_basic_aggrgte_day	Daily	D-2	0	534879		530559	530484	530422	530349	530082	529859
solid_scb_dpst_acct_acct_basic_aggrgte_day	Daily	D-2	0	23300130		23153078	23146828	23142087	23136392	23127187	23118874
solid_scb_dpst_acct_cust_basic_aggrgte_day	Daily	D-2	0	12160115		12120781	12118800	12117801		12112932	12109907

data

As of

ตุลาคม 2023

Overall DQI Score

89.0904%

(↑ 0.4393%)

Total # of CDEs

14

No. of DQ Rules

53

Data Subject Area

☐ Customer Profile

☐ Deposit

☐ Secured Loan

Completeness

99.9603%

(↑ 0.0007%)

Conformity

97.9582%

(↑ 0.6605%)

Integrity

(Blank)

N/A

Uniqueness

99.9875%

(↑ 0.0226%)

Consistency

0.0665%

(↓ -66.4508%)

Accuracy

58.0916%

(↑ 0.6841%)

Calculation Method

Current

MOM%

Filter Threshold

ALL

Good

Fair

Poor

Rule Type

Error

Warning

By Data Set

By Data Element

Dataset

☐ CORE_DATAX_CUST_PRFL_CUS...

☐ CORE_SCB_DPST_TXN_ACCT_B...

☐ CORE_SCB_DPST_TXN_ACCT_B...

☐ CORE_SCB_DPST_TXN_ACCT_B...

☐ SOLID_DATAX_CUST_PRFL_CUS...

☐ SOLID_SCB_AUTO_ACCT_ACCT...

☐ SOLID_SCB_AUTO_ACCT_CUST...

☐ SOLID_SCB_DPST_ACCT_ACCT...

☐ SOLID_SCB_DPST_ACCT_CUST...

Data element name

DQI score

Completeness

Conformity

Integrity

Uniqueness

Consistency

Accuracy

Active and Dormant Individual All Deposit Account Type Average Month-to-date Account Amount

56.3458%

56.3458%

Active and Dormant Individual Saving Deposit Account Type Last Inflow Transaction Date List

88.0392%

77.8450%

100.0000%

Active and Dormant Individual Saving Deposit Account Type Total Current Balance Percentile Ranking Number

100.0000%

100.0000%

Latest Customer Monthly Income Amount

0.6559%

0.6559%

Latest Release Collateral Amount

53.1077%

53.1077%

Latest Release Collateral Date

81.0114%

81.0114%

Latest Release Collateral List

80.8954%

100.0000%

53.0527%

Total Active Auto Loan Accounts Count

66.6667%

100.0000%

0.0000%

Total Active Auto Loan Outstanding Balance Amount

100.0000%

100.0000%

100.0000%

Total Active Auto Loan Red Book Amount

0.0000%

0.0000%

Total Auto Loan Applications Approved Count

99.2865%

98.5730%

100.0000%

Total Auto Loan Applications Count

95.3392%

91.1094%

99.9817%

Total Closed Auto Loan Accounts Count

50.0000%

0.0166%

99.9834%

89.0904%

99.9603%

97.9582%

99.9875%

0.0665%

58.0916%

GLOBAL THRESHOLD

GOOD

FAIR

POOR

>98%

95-98%

<95%

Rule ID

Business rule description

Good

Fair

Poor

SOLID_SCB_AUTO_ACCT_CUST_BASIC_AGGRGTE_DAY.FIN_SCB_AUTO_TOT_ACCT_HLDNG_ACTV_CNT_DLY_V1_CONSISTENCY_1

Active Count = Total - Closed

>98%

95-98%

<95%

SOLID_SCB_AUTO_ACCT_CUST_BASIC_AGGRGTE_DAY.FIN_SCB_AUTO_TOT_APPL_APRV_CNT_DLY_V1_ACCURACY_1

Approved <= Total application

>98%

95-98%

<95%

SOLID_SCB_AUTO_ACCT_CUST_BASIC_AGGRGTE_DAY.FIN_SCB_AUTO_TOT_ACCT

At least 1 active account exists

>98%

95-98%

<95%

Data Governance and Catalog

// internal //

CardX

As of
กันยายน 2023

Overall DQI Score

91.7666%
N/A

Total # of CDEs

20

No. of DQ Rules

46

Data Subject Area

Credit Card and Speedy Cash Information

Completeness

88.8044%
N/A

Conformity

90.3603%
N/A

Integrity

100.0000%
N/A

Uniqueness

99.9998%
N/A

Consistency

99.9998%
N/A

Accuracy

85.0284%
N/A

Calculation Method

Current

MOM%

Filter Threshold

ALL

Good

Fair

Poor

Rule Type

Error

Warning

By Data Set

By Data Element

Dataset

CRCARD_SPC_LN_ORIG

Dataset name

DQI Score

Completeness

Conformity

Integrity

Uniqueness

Consistency

Accuracy

CRCARD_SPC_LN_ORIG

91.7666%

88.8044%

90.3603%

100.0000%

99.9998%

99.9998%

85.0284%

Loan Application Status Timestamp

100.0000%

100.0000%

100.0000%

Loan Application Status Code

100.0000%

100.0000%

100.0000%

100.0000%

Loan Application Rejected Timestamp

100.0000%

100.0000%

100.0000%

Loan Application Registration Timestamp

100.0000%

100.0000%

100.0000%

Loan Application Registration Identifier

72.4180%

100.0000%

17.2541%

99.9998%

Loan Application Product Type Code

100.0000%

100.0000%

100.0000%

100.0000%

CRCARD_SPC_LN_ORIG.PROD_CD_INTEGRITY_1

100.0000%

100.0000%

CRCARD_SPC_LN_ORIG.PROD_CD_CONFORMITY_1

100.0000%

100.0000%

CRCARD_SPC_LN_ORIG.PROD_CD_COMPLETENESS_1

100.0000%

100.0000%

Loan Application Approval Timestamp

100.0000%

100.0000%

100.0000%

Loan Application Approval Reason Code

99.9989%

100.0000%

100.0000%

99.9952%

CRCARD_SPC_LN_ORIG.APRV_RSN_CD_INTEGRITY_1

100.0000%

100.0000%

91.7666%

88.8044%

90.3603%

100.0000%

99.9998%

99.9998%

85.0284%

GLOBAL THRESHOLD

GOOD

FAIR

POOR

>98%

95-98%

<95%

Rule ID

Business rule description

Good

Fair

Poor

CRCARD_SPC_LN_ORIG.RGTR_ID_CONFORMITY_1

Application Identifier should follow a specified alphanumeric format, starts with 2 letters then follows with numeric

>98%

95-98%

<95%

CRCARD_SPC_LN_ORIG.RGTR_DT_ACCURACY_1

Application Registration Date/Time cannot be defined as a future date

>98%

95-98%

<95%

CRCARD_SPC_LN_ORIG.BRTH_YR_CONFORMITY_1

Birth Date should follow a specified year format

>98%

95-98%

<95%

CRCARD_SPC_LN_ORIG.CUST_NUM_CONSISTENCY_1

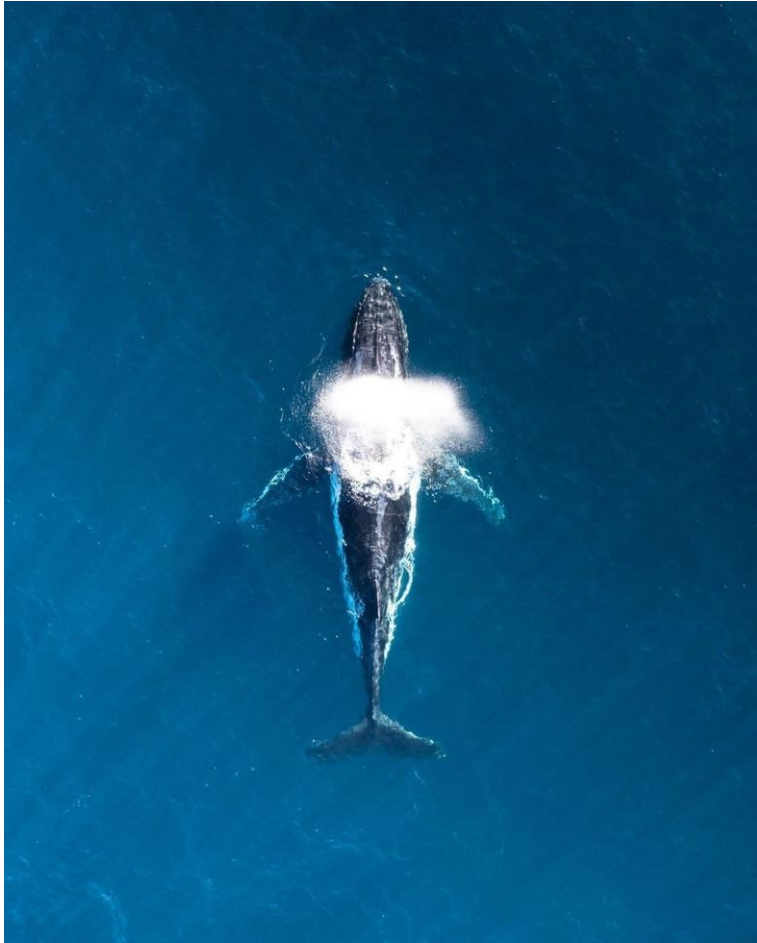
Customer Number in CRCARD_SPC_LN_ORIG should correspond to Customer Number in

>98%

95-98%

<95%

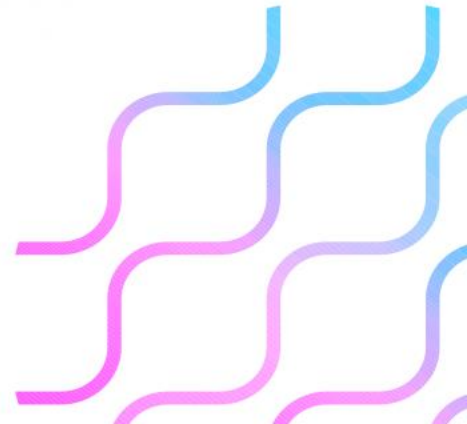
Data Governance and Catalog



03

Remediate Phase

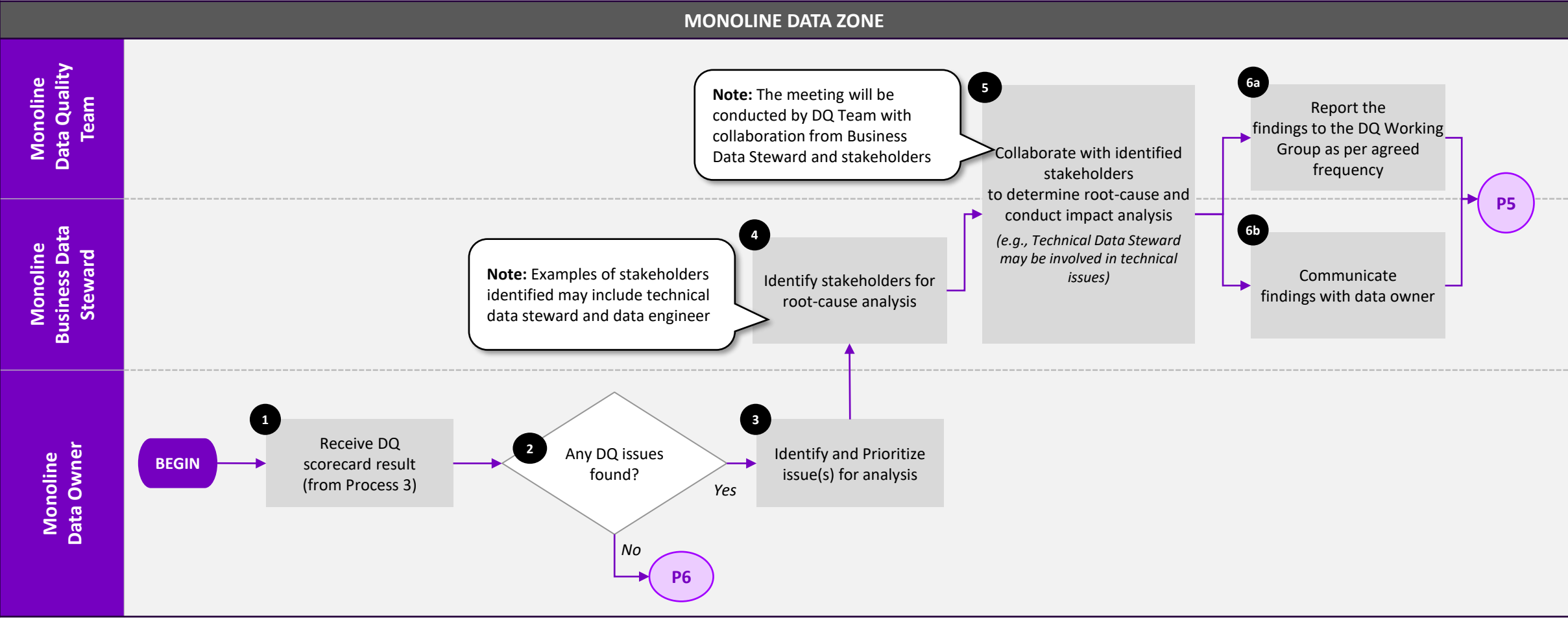
Prioritize issues and conduct root-cause and impact analysis for DQ issue



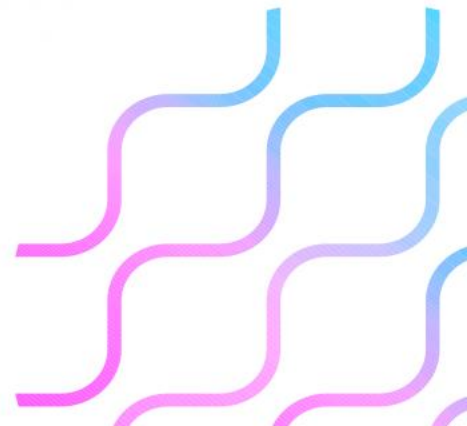
PROCESS 4 (MDZ) TO PRIORITIZE AND ANALYZE DQ ISSUES IDENTIFIED

Prerequisite(s)

- Scorecard analysis is also done (in Process 3)
- Issue is identified and added to data owner’s backlog



Sample case : DQ issue Root-cause Analysis



DataX GDZ Data Quality Issue : Root-cause analysis and proposed improvement action

Key Issue: GDZ-C360 Team found inconsistent data from scb cust_info to identify scb staff record

Assumption: The record of customer who is SCB staff will always have "crm proxy income" = 0

Product: C360 Area: Customer Profile

Zone: GDZ Data Set: solid_customer_profile

Raised Date: 17 Mar 23

Product owner: K.Ekularn

Priority: Low

Issue assessment

Data Quality



DQ Accuracy Rule

- C360 team require to validate data element name: **Latest Customer Monthly Income**, in GDZ by define DQ business Rule as following:
DQ Rule Condition: Latest Customer Monthly Income Amount in dataset solid_customer_profile must more that 0
- After execute there are the result failed - found Latest Customer Monthly Income = 0
- C360 and DQ team investigated under assumption that,
if the customer is scb employee, the crm_prxy_incm will be masking as zero.

Issue findings

C360 Product Owner & FA



SCB Datalake

C360 team explored scb datalake **Dataset name:** dlprod_curated_cust_db.cust_info, with is the source used, found that:

- There are 2 column* : **scb_stff an stff_flag**, which the value can be different due to the definition. See note below.
- **If scb_stff = "Y", it is the record of scb or scbx staff, then crm_prxy_incm should be 0**
- From assumption above , the issue raised by the query from dataset found as below :

scb_stff	stff_flag	total_rows	total_null_crm_prxy_incm	total_0_crm_prxy_incm	total_over0_crm_prxy_incm
N	N	32218346	715625	319	31502402
N	Y	3947	0	3900	47
Y	Y	20525	0	20448	77

Found scb staff= 77 records with income > 0

Why? Staff record in dataset: cust_info contain crm_prxy_incm > 0 ?

Informed by FA:
scb_stff: Y = SCB and SCBX staff
stff_flag: Y = SCB temporary staff

DataX GDZ Data Quality Issue : Root-cause analysis and proposed improvement action

Key Issue: GDZ-C360 Team found inconsistent data from scb cust_info

Product: C360

Area: Customer Profile

Zone: GDZ

Data Set: solid_customer_profile

Raised Date: 17 Mar 23

Product owner: K.Ekularn

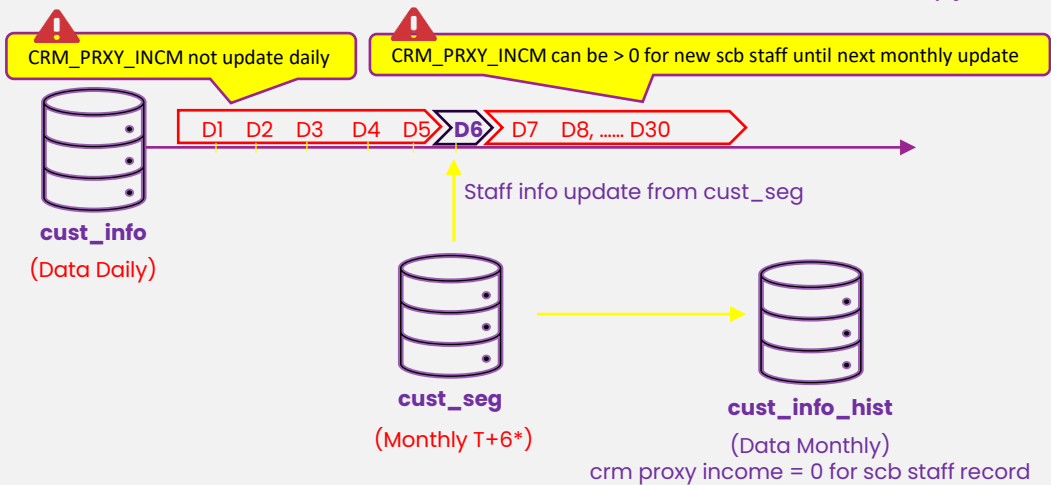
Priority: Low

Impact

- Misleading data usage
- Risk to employees PII data leak

Root cause

- By designed, table: cust_info is a daily job will contain actual information of customer record until the monthly job which will get the scb staff updated from table: cust_seg where crm_proxy_income of scb staff already masked(amount=0)
- However, even table: cust_info have updated of scb staff on the monthly basis, new more record of scb staff would added with actual data until next monthly job.



Recommendation

- To avoid PII leakage, recommend to use table: cust_info_hist which crm proxy income of scb staff already masking (amount=0). However, table: cust_info_hist is monthly table (T+6)

```
1 SELECT dl_data_dt,rm_key, scb_stff, stff_flag,crm_prxy_incm
2 FROM dlprod_curated_cust_db.cust_info_hist
3 WHERE scb_stff = 'Y'
4 AND stff_flag = 'Y'
5 AND crm_prxy_incm > 0
6 and dl_data_dt in ('2023-05-31','2023-04-30')
7
```

► (4) Spark Jobs
Query returned no results

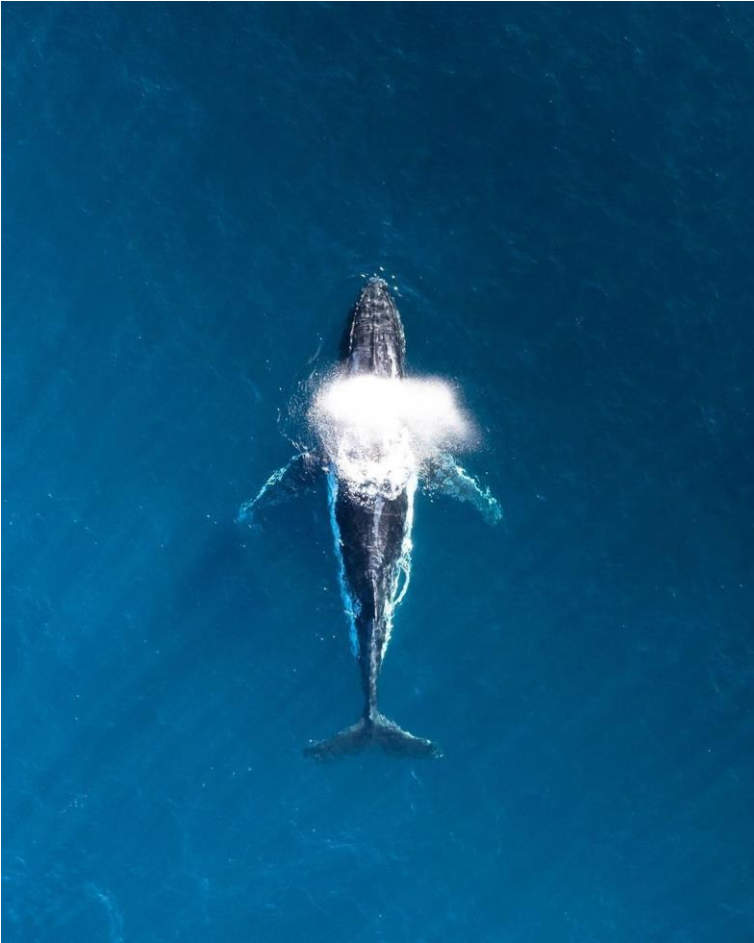
- If product need to run on daily basis using table: cust_info, so the risk may need to define the mitigation by product owner.

Action

- Discuss with C360 Product team for risk mitigation
- Inform scb DGO to recheck business definition of staff flag and temporary staff

Conclusion

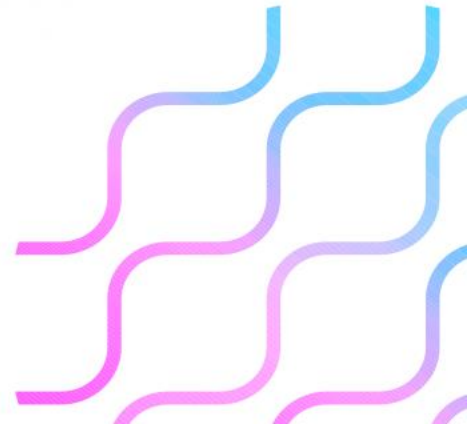
C360 Product owner insists the need to use table cust_info for daily tracking change purpose. However, CRM proxy income can no longer be shared from SCB since it is part of exclusion item.



04

Monitor Phase

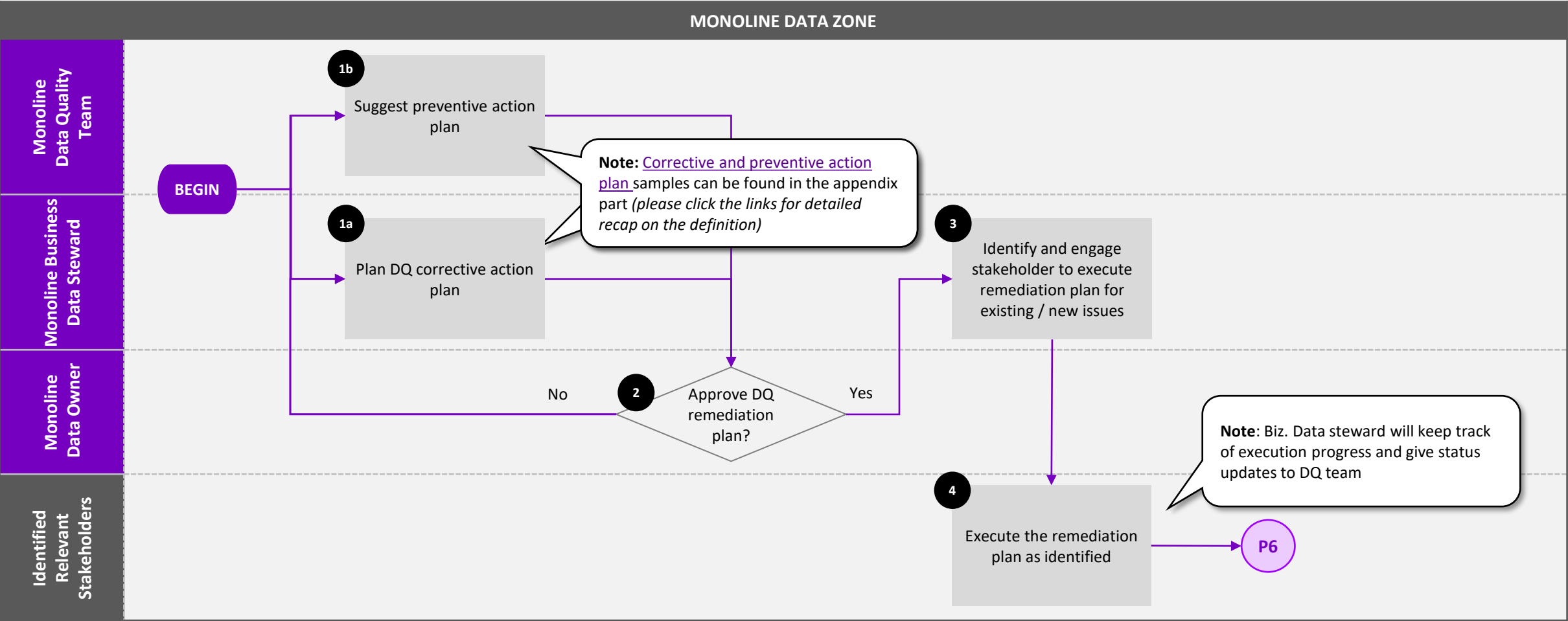
DQ Issue Remediation & DQ Scorecard Monitoring



PROCESS 5 (MDZ) TO DEVELOP, EXECUTE, AND MONITOR REMEDIATION PLAN

Prerequisite(s)

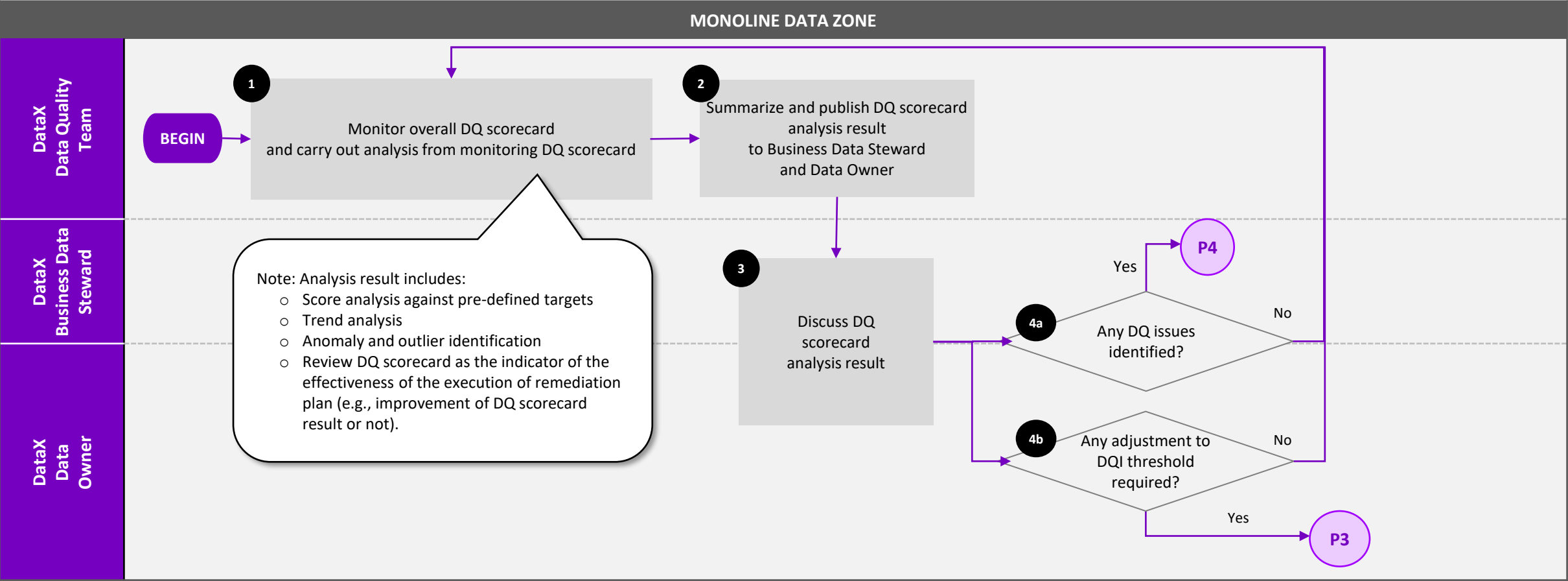
- Data Quality issues in MDZ are analyzed



PROCESS 6 (MDZ) TO MONITOR AND COMMUNICATE THE DATA QUALITY SCORECARD RESULTS

Prerequisite(s)

- Data Quality Scorecard is generated
- Data Quality Tool / Scorecard is established and updated



Activities and Plan



✓ DQ Operating shape up

Expectation and what we will achieve



✓ Define and Discover Phase

Introduce Business rule with generic rule automation for CDEs



✓ Diagnose Phase

Conduct Business rules refinement
Illustrate DQ results design and DQ scorecard



✓ Remediate and monitor Phase

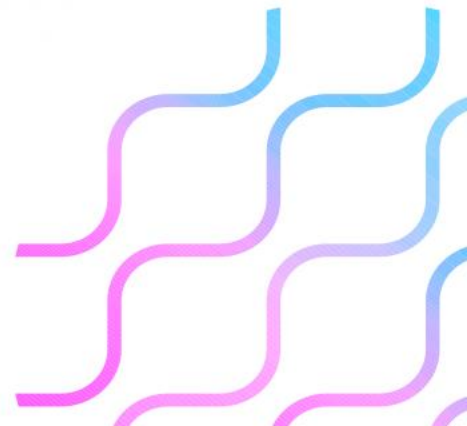
Recap DQ Remediation Process



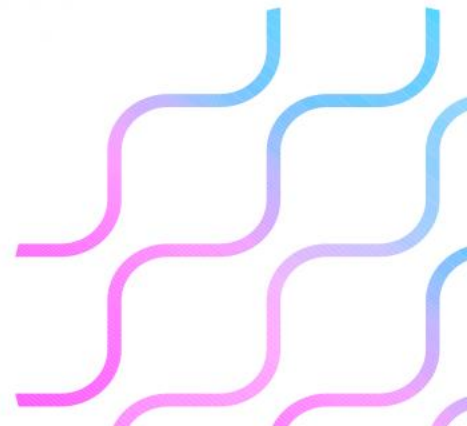
Q&A

If any

Thank you



Appendix



PROCESS 1 (MDZ) TO IDENTIFY/MAINTAIN CDE AND MAP TO DQ DIMENSION: DETAILED ACTIVITIES

MONOLINE DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Identify/maintain whether the data elements fall under CDEs, based on CDE criteria as defined by DataX Data Quality Team	Monoline Data Quality (DQ) Team; Monoline Business Data Steward	Monoline DQ Team and Business Data Steward to decide whether the data elements are Critical Data Elements (CDEs) or not, as per the CDE criteria defined by DataX
1+	(Optional) Prioritize CDEs with defined approach	Monoline DQ Team; Monoline Business Data Steward	Monoline DQ Team and Business Data Steward to decide, among identified CDEs, which to be prioritized, as per the approach defined by DataX
2	CDE* or not?	Monoline DQ Team; Monoline Business Data Steward	Monoline DQ Team and Business Data Steward to decide whether the data element is (prioritized) CDE or not: • If yes, proceed to step 3 • If not, process ends
3	Map CDE* to DQ dimension for preliminary assessment	Monoline DQ Team	Monoline Data Quality Team to map (prioritized) CDE to DQ dimension by relevancy for preliminary assessment on data quality
4	Review mapped DQ dimension & collect feedback with related stakeholders	Monoline DQ Team; Monoline Business Data Steward	Monoline DQ Team to review mapped DQ dimension with Monoline Business Data Steward and collect feedback with related stakeholders

* If the optional step of prioritization is adopted, the CDE in Step 2 and 3 will refer to the prioritized CDE

PROCESS 1 (MDZ) PROCESS TO IDENTIFY/MAINTAIN CDE AND MAP TO DQ DIMENSION: RACI MAP

RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key Activity	MONOLINE DATA ZONE				
		Monoline Data Quality (DQ) Team		Monoline Biz. Data Steward	Monoline Data Owner	Monoline Technical Data Steward
		DQ Officer	Head of DQ			
1	Identify/maintain whether the data elements fall under CDEs, based on CDE criteria as defined by DataX Data Quality Team	R	A	R	A	I
1+	(Optional) Prioritize CDEs with defined approach	R	A	R	A	I
2	CDE* or not?	R	A	R	A	I
3	Map CDE* to DQ dimension for preliminary assessment	R	A	C	C	I
4	Review mapped DQ dimension & collect feedback with related stakeholders	R	A	R	A	C

* If the optional step of prioritization is adopted, the CDE in Step 2 and 3 will refer to the prioritized CDE

PROCESS 2 (MDZ) TO IDENTIFY AND REFINE DATA QUALITY BUSINESS RULE: DETAILED ACTIVITIES

MONOLINE DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Initiate and complete data profiling to understand the characteristics and relationship of the data	Monoline DQ Team	Monoline Data Quality Team to run data profiling to gain understanding of the characteristics and relationship of the data
2	Review and discuss findings from data profiling	Monoline DQ Team; Monoline Business Data Steward	Monoline Data Quality Team and Business Data Steward to review and discuss the findings from data profiling
3	Draft/ maintain fundamental Data Quality (DQ) business rules in the template	Monoline DQ Team	Monoline Data Quality Team to develop and provide fundamental DQ business rules
4	Refine and finalize the full list of DQ business rules	Monoline Business Data Steward	Monoline Business Data Steward to refine the fundamental DQ business rules prepared by DQ team and develop/update the final full list of DQ business rules
5	Review the finalized full list of DQ business rules prepared by Biz. Data Steward	Monoline Data Owner	Monoline Data Owner to review the finalized full list of DQ business rules developed/updated by the Business Data Steward
6	Approve the final full list of DQ business rules?	Monoline Data Owner	Monoline Data Owner to decide whether to approve the finalized full list of DQ business rules: • If yes, proceed to step 7 • If not, proceed to step 4 (i.e., Monoline Biz. Data Steward to update the full list)
7	Review the approved full list	Monoline DQ Team; Monoline Business Data Steward	Monoline Data Quality Team and Business Data Steward to review the final full list of DQ business rules
8	Update the final full list in the DQ business rules template	Monoline DQ Team	Monoline DQ Team to update the finalized full list approved by the Data Owner in the DQ business rules template

PROCESS 2 (MDZ) TO IDENTIFY AND REFINE DATA QUALITY BUSINESS RULE: RACI MAP

RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key activity	MONOLINE DATA ZONE				
		Monoline Data Quality (DQ) Team		Monoline Biz. Data Steward	Monoline Data Owner	Monoline Technical Data Steward
		DQ Officer	Head of DQ			
1	Initiate and complete data profiling to understand the characteristics and relationship of the data	R	A	C	C	C
2	Review and discuss findings from data profiling	R	A	R	A	I
3	Draft/ maintain fundamental Data Quality (DQ) business rules in the template	R	A	I	I	I
4	Refine and finalize the full list of DQ business rules	C	I	R	I	C
5	Review the finalized full list of DQ business rules prepared by Biz. Data Steward	C	I	C	R, A	I
6	Approve the final full list of DQ business rules?	I	I	I	R, A	I
7	Review the approved full list	R	A	R	A	I
8	Update the final full list in the DQ business rules template	R	A	I	I	C

PROCESS 3 (MDZ) TO GENERATE AND ANALYZE DQ SCORECARD: DETAILED ACTIVITIES

GROUP DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Generate and run DQ business rules	Monoline Data Quality (DQ) Team	Monoline DQ Team to apply and run DQ business rule and generate the DQ scorecard via the DQ tool once the DQ business rules are implemented and the DQ Scorecard is built. Additionally, the timeliness dimension will be used to measure timeliness at Data Set level.
2	Consolidate overall DQI performance result into DQ scorecard generated	Monoline DQ Team	After DQ scorecard is generated, Monoline DQ Team to consolidate the overall result from DQ scorecard
3	Carry out analysis result of DQ scorecard	Monoline DQ Team, Monoline Data Owner	Monoline DQ team to analyze the result of the DQ scorecard generated with the following methods (non-exhaustive): - Score analysis against pre-defined targets - Trend analysis - Anomaly and outlier identification
4	Is an adjustment to Target DQI Threshold required?	Monoline Data Owner	Monoline Data Owner to decide if an adjustment to the Target DQI threshold is required by considering the CDE classification, context and use case of the data - If yes, proceed to Step 5 - If no, proceed to Process 4 (to identify whether there is any DQ issue)
5	Adjust target DQI Threshold for that selected CDE	Monoline Data Owner	If a target DQI threshold is required, Monoline Data Owner to determine the DQI threshold to be applied and adjust the Target DQI threshold accordingly

PROCESS 3 (MDZ) TO GENERATE AND ANALYZE DQ SCORECARD: RACI MAP

RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key Activity	MONOLINE DATA ZONE				
		Monoline Data Quality (DQ) Team		Monoline Biz. Data Steward	Monoline Data Owner	Monoline Technical Data Steward
		DQ Officer	Head of DQ			
1	Generate and run DQ business rules	R	A	I	C	I
2	Consolidate overall DQI performance result into DQ scorecard generated	R	A	I	C	I
3	Carry out analysis result of DQ scorecard	R	A	I	R	I
4	Is an adjustment to Target DQI Threshold required?	C	A	C, I	R	C, I
5	Adjust target DQI Threshold for that selected CDE	C	A	C, I	R	C, I

PROCESS 4 (MDZ) TO PRIORITIZE AND ANALYZE DQ ISSUES IDENTIFIED: DETAILED ACTIVITIES

MONOLINE DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Receive DQ scorecard result (from Process 3)	Monoline Data Owner	Monoline Data Owner to receive DQ scorecard result from Process 3
2	Any DQ issues found?	Monoline Data Owner	If no DQ issues found, Monoline Data Owner to proceed to Process 6 If DQ issues found from DQ scorecard, Monoline Data Owner to proceed to step 3
3	Identify and Prioritize issue(s) for analysis	Monoline Data Owner	Monoline Data Owner to determine prioritized issue(s) for analysis based on the list of issues in the backlog and CDE classification of the impacted data
4	Identify stakeholders for root-cause analysis	Monoline Business Data Steward	Monoline Business Steward to identify stakeholders required to investigate the issue. Stakeholders may include data engineer, technical data steward etc.
5	Collaborate with identified stakeholders to determine root-cause and conduct impact analysis <i>(e.g., Technical Data Steward may be involved in technical issues)</i>	Monoline Business Steward; Monoline DQ Team	Monoline DQ Team to conduct a joint meeting for Monoline Business Steward and Monoline Data Quality Team to jointly drive discussions to drill-down on DQ issues to identify how and where the issue stemmed from, and analyze its impact
6a	Report the findings to the DQ Working Group as per agreed frequency	Monoline DQ Team	Monoline Data Quality Team to report the findings from root-cause analysis and impact analysis to the DQ Working Group as per agreed frequency
6b	Communicate findings with data owner	Monoline Business Data Steward	Monoline Data Steward to communicate findings from root-cause analysis and impact analysis with Data Owner

PROCESS 4 (MDZ) TO PRIORITIZE AND ANALYZE DQ ISSUES IDENTIFIED: RACI MAP

RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key Activity	Monoline DATA ZONE				
		Monoline Data Quality (DQ) Team		Monoline Biz. Data Steward	Monoline Data Owner	Monoline Technical Data Steward
		DQ Officer	Head of DQ			
1	Receive DQ scorecard result (from Process 3)	C	C	I	A, R	I
2	Any DQ issues found?	C, I	C, I	I	A, R	I
3	Identify and Prioritize issue(s) for analysis	C, I	C, I	I	A, R	I
4	Identify stakeholders for root-cause analysis	I	I	R	A, I	C
5	Collaborate with identified stakeholders to determine root-cause and conduct impact analysis (e.g., Technical Data Steward may be involved in technical issues)	R	I	R	A, I	C
6a	Report the findings to the DQ Working Group as per agreed frequency	R	A	I	I	I
6b	Communicate findings with data owner	I	I	R	A	I

PROCESS 5 (MDZ) TO DEVELOP, EXECUTE, AND MONITOR REMEDIATION PLAN: DETAILED ACTIVITIES

MONOLINE DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1a	Plan DQ corrective action plan	Monoline Business Data Steward	While Monoline DQ Team is suggesting preventive action plan, Monoline Business Data Steward to plan corrective action plan and provide the information in governance forums if it is required
1b	Suggest preventive action plan	Monoline Data Quality (DQ) Team	Monoline DQ Team to suggest preventive action plan to solve data quality issue
2	Approve DQ remediation plan?	Monoline Data Owner	Monoline Data Owner to approve the DQ remediation plan provided by Monoline Business Data Steward and Monoline DQ Team. If the corrective action plan is not approved, go back to step 1a. If preventive action plan is not approved, go back to step 1b If both corrective and preventive plan are approved, proceed to step 3.
3	Identify and engage stakeholder to execute remediation plan for existing / new issues	Monoline Business Data Steward	Monoline Business Data Steward to identify the relevant stakeholder for remediation plan (both preventive action and corrective action plan) and engage them to execute the plan
4	Execute the remediation plan as identified	Identified Relevant Stakeholder	Identified Relevant Stakeholder to execute the remediation to solve the data quality issue

PROCESS 5 (MDZ) TO MONITOR AND COMMUNICATE THE DATA QUALITY SCORECARD RESULTS : RACI MAP

RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key activity	MONOLINE DATA ZONE					
		DataX Data Quality (DQ) Team		DataX Biz. Data Steward	DataX Data Owner	DataX Technical Data Steward	Identified Stakeholder
		DQ Officer	Head of DQ				
1a	Plan DQ corrective action plan	I	I	R	A	C	-
1b	Suggest preventive action plan	R	A	I	I	C	-
2	Approve DQ corrective action plan?	I	I	I	R, A	I	-
3	Identify and engage stakeholder to execute remediation plan for existing / new issues	C	C	R	A	C	I
4	Execute the remediation plan as identified	I	I	C	C	C	R, A

PROCESS 6 (MDZ) TO MONITOR AND COMMUNICATE THE DATA QUALITY SCORECARD RESULTS: DETAILED ACTIVITIES

MONOLINE DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Monitor overall DQ scorecard, and carry out analysis from monitoring DQ scorecard	Monoline Data Quality (DQ) Team	Monoline DQ Team to monitor DQ scorecard periodically as frequency defined in metrics and carry out analysis on the result of DQ scorecard from monitoring
2	Summarize and publish DQ scorecard analysis result to Business Data Steward and Data Owner	Monoline DQ Team	Monoline DQ Team to summarize and publish the DQ scorecard result to Monoline Business Data Steward and Data Owner for their further discussion and analysis
3	Discuss DQ scorecard analysis result	Monoline Biz. Data Steward; Monoline Data Owner	Monoline Business Data Steward and Data Owner to discuss the analysis result of DQ scorecard
4a	Any DQ issues identified?	Monoline Biz. Data Steward; Monoline Data Owner	Monoline Business Data Steward and Data Owner to identify whether any DQ issues need to be solved • If DQ issue is identified, proceed to Process 4 • If no DQ issue is identified, the process will continue back to Step 1
4b	Any adjustment to DQI threshold required?	DataX Data Owner	DataX Data Owner to identify whether any systematic adjustment are needed: • If adjustment is needed for DQI threshold, proceed to Process 3 • If no adjustment is needed, the process will continue back to Step 1

PROCESS 6 (MDZ) TO MONITOR AND COMMUNICATE THE DATA QUALITY SCORECARD RESULTS : RACI MAP

RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key activity	MONOLINE DATA ZONE				
		Monoline Data Quality (DQ) Team		Monoline Biz. Data Steward	Monoline Data Owner	Monoline Technical Data Steward
		DQ Officer	Head of DQ			
1	Monitor overall DQ scorecard, and carry out analysis from monitoring DQ scorecard	R	A	I	I	I
2	Summarize and publish DQ scorecard analysis result to Business Data Steward and Data Owner	R	A	I	I	I
3	Discuss DQ scorecard analysis result	C	C	R	A	I
4a	Any DQ issues identified?	C	C	R	A	I
4b	Any adjustment to DQI threshold required?	C	C	I	R, A	I